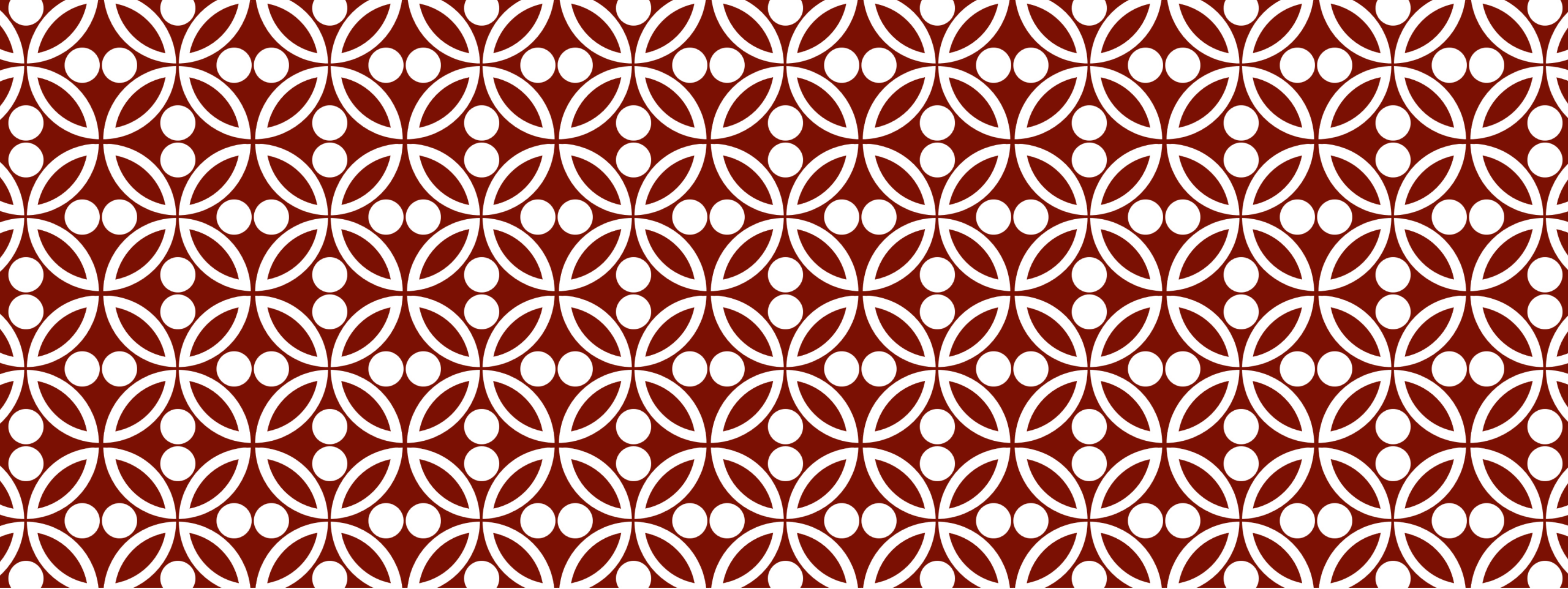


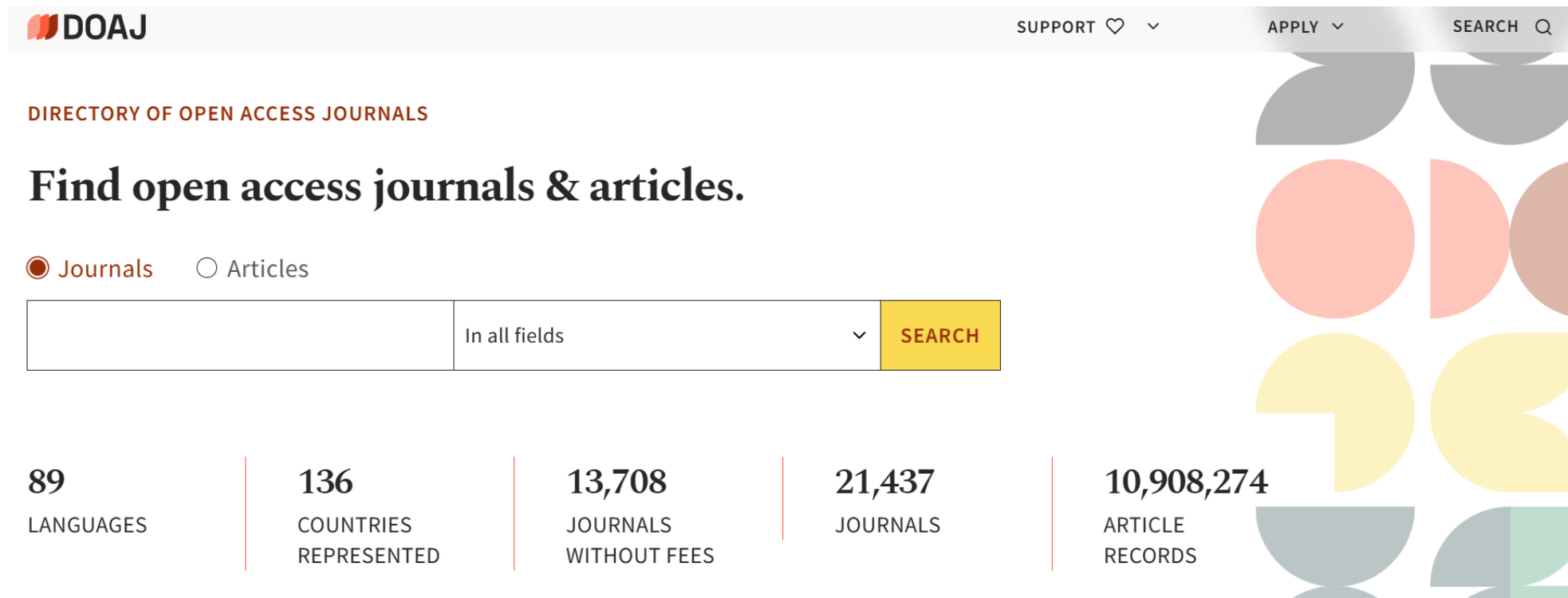


**DEEP LEARNING FOR ANALYSIS OF ARTICLES RELATED TO
TAXATION, ECONOMIC GROWTH, AND TECHNOLOGY**

Lina Zadorozhnia

Summary

The project aims to create a neural network in order to find and analyze with machine learning techniques academic papers from DOAJ (**Directory of Open Access Journals**) which are written in English and related to the topics of taxation, economic growth, and technology. Particularly, based on the abstract, train the model to predict keywords.



DOAJ

SUPPORT  

APPLY 

SEARCH 

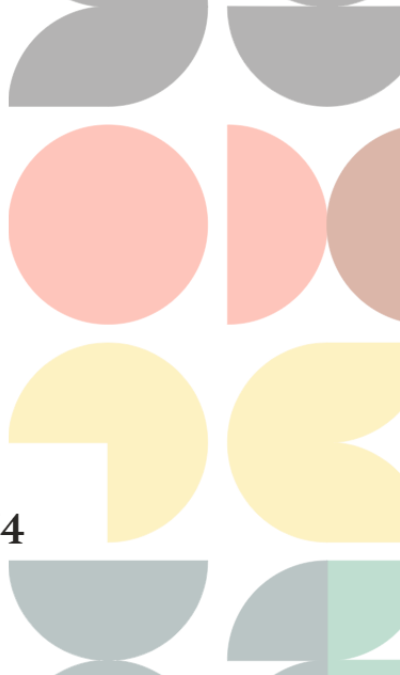
DIRECTORY OF OPEN ACCESS JOURNALS

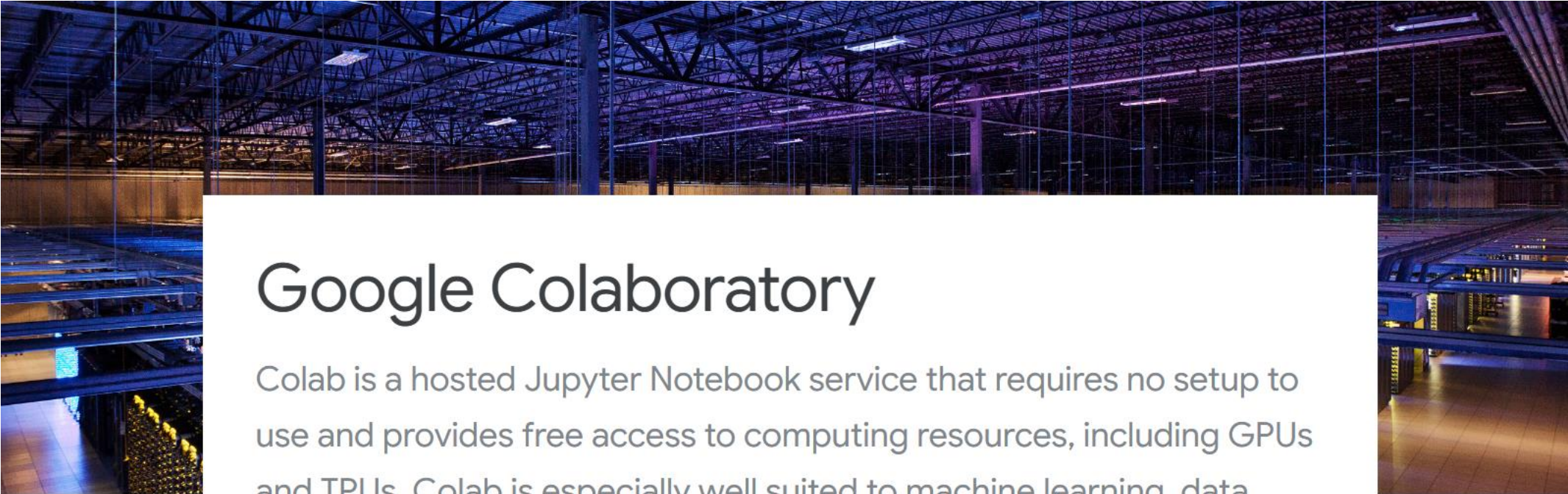
Find open access journals & articles.

☒ Journals ☐ Articles

In all fields  **SEARCH**

89 LANGUAGES	136 COUNTRIES REPRESENTED	13,708 JOURNALS WITHOUT FEES	21,437 JOURNALS	10,908,274 ARTICLE RECORDS
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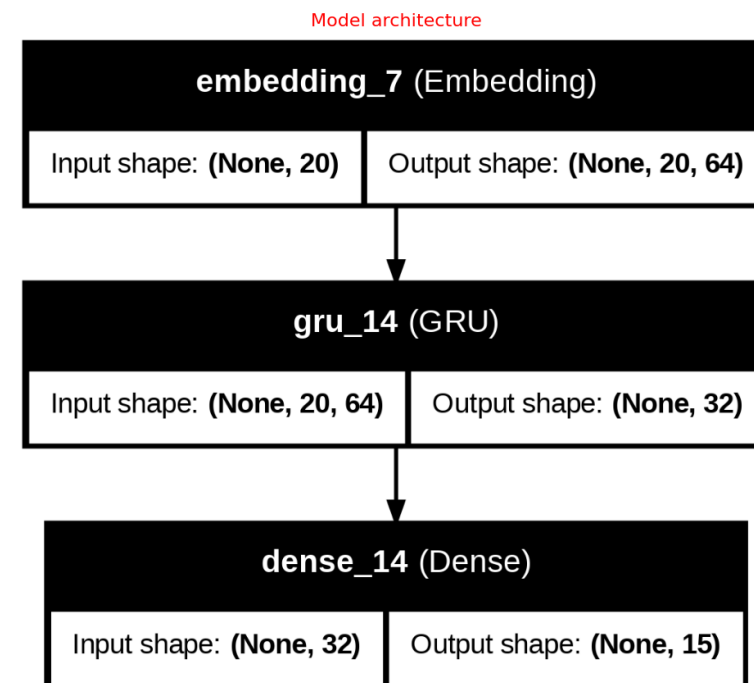
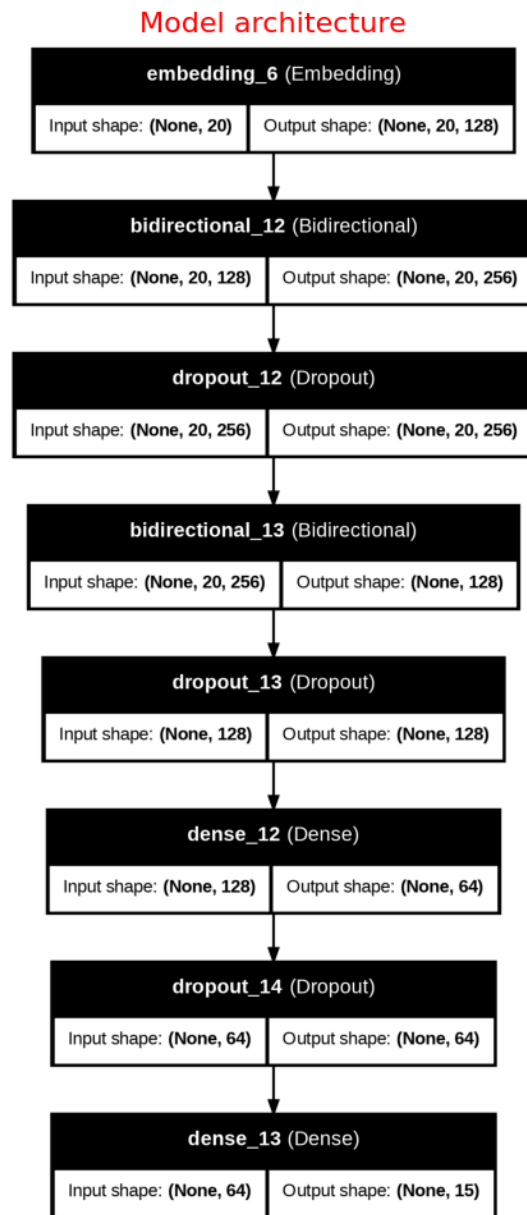
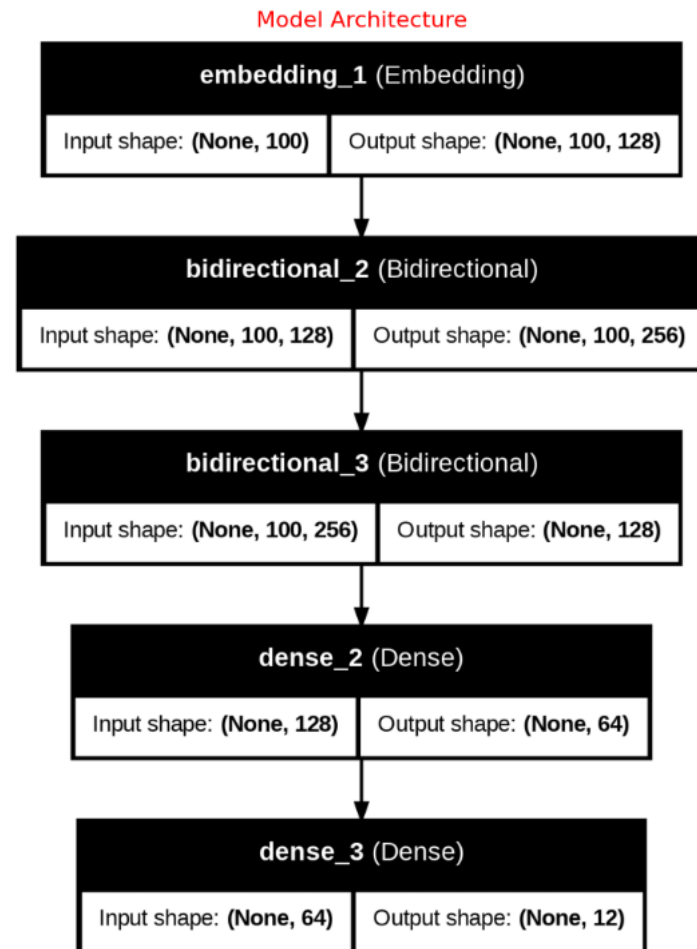


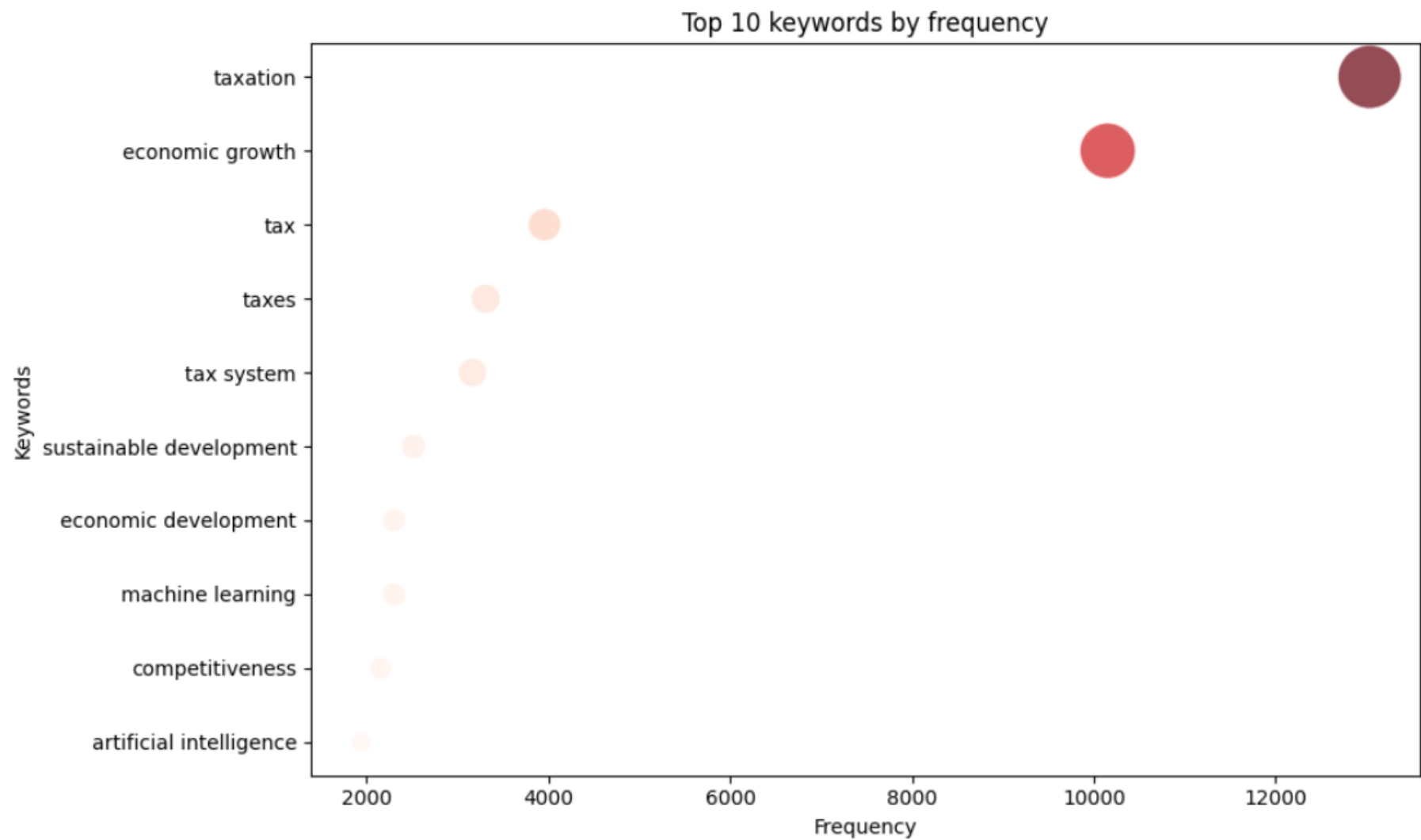


Google Colaboratory

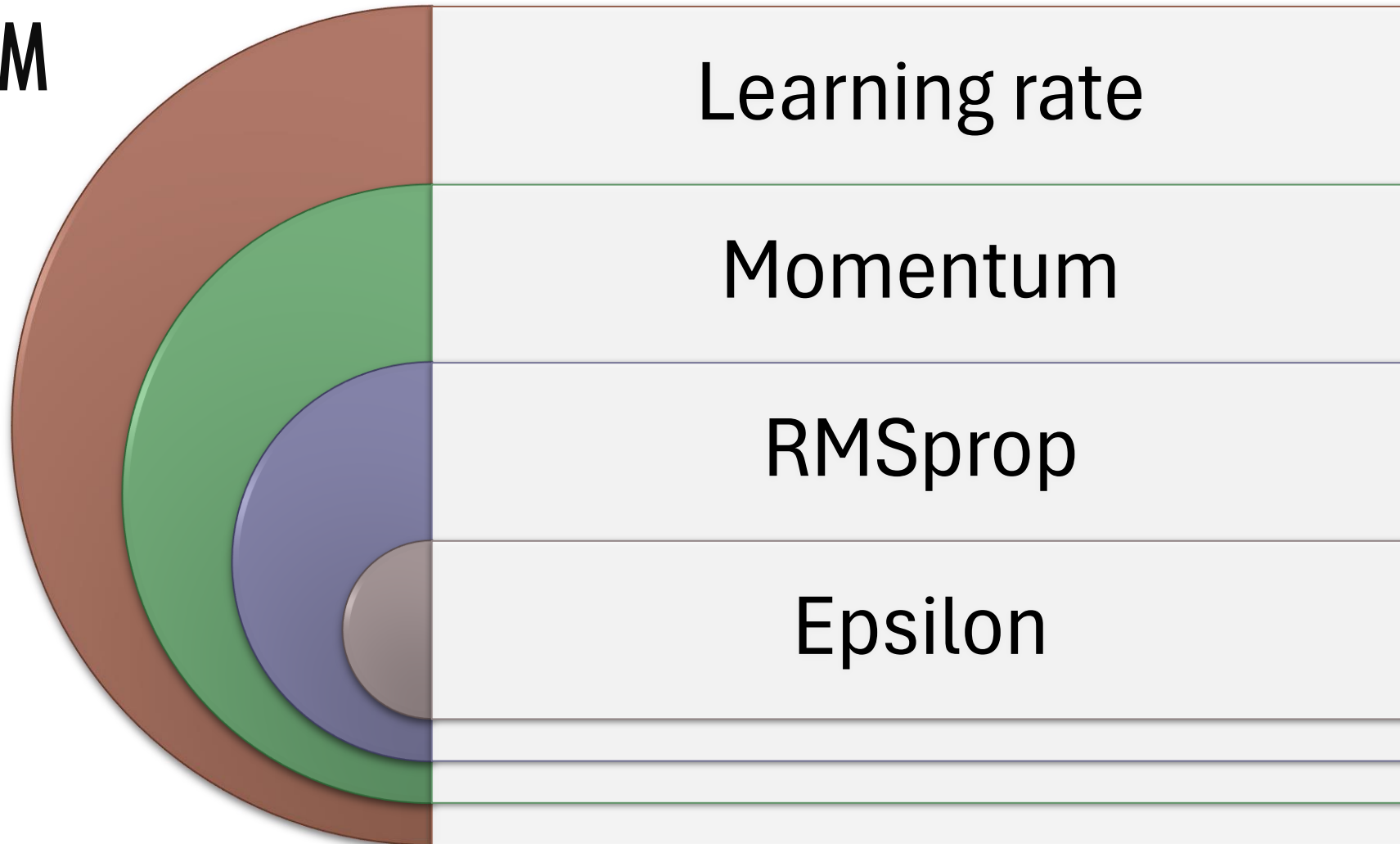
Colab is a hosted Jupyter Notebook service that requires no setup to use and provides free access to computing resources, including GPUs and TPUs. Colab is especially well suited to machine learning, data science, and education.

[Open Colab](#)[New Notebook](#)





ADAM



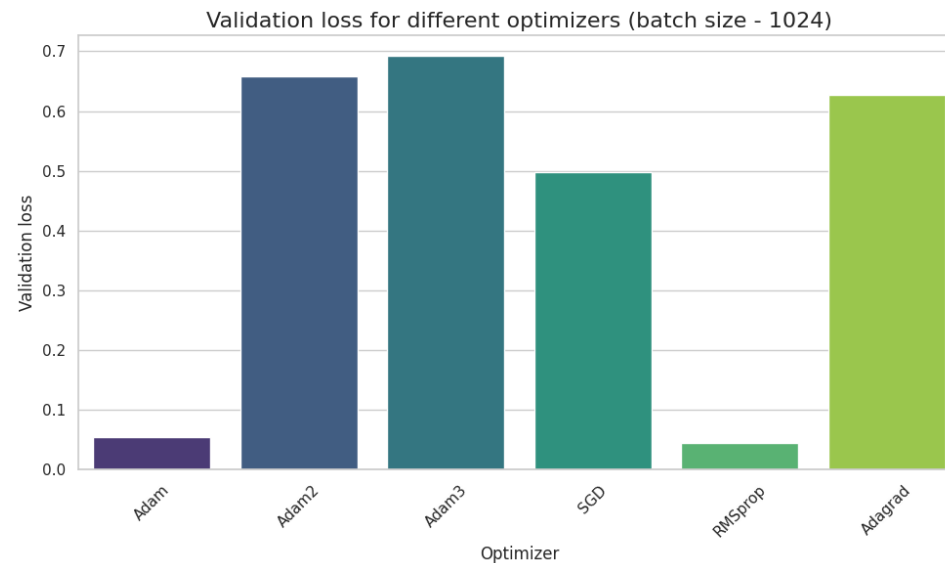
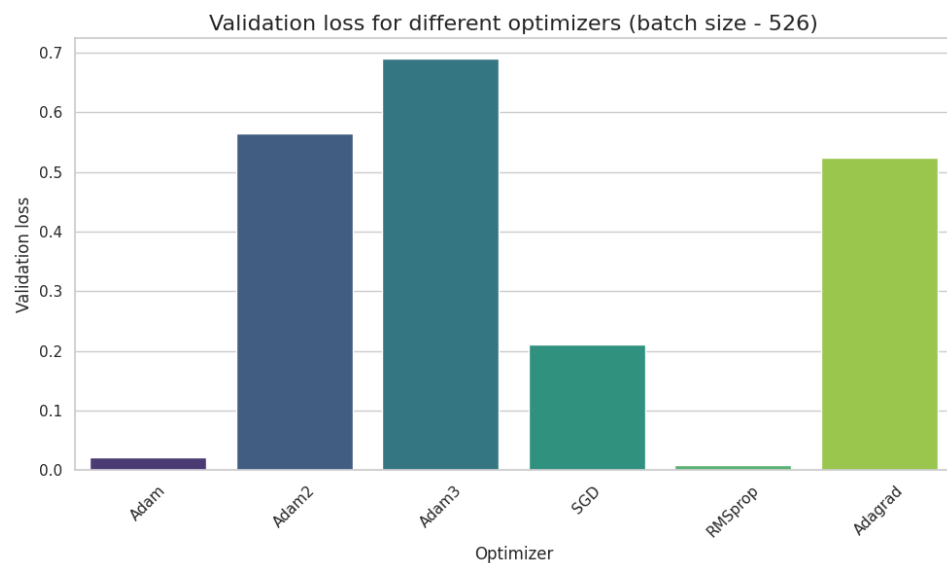
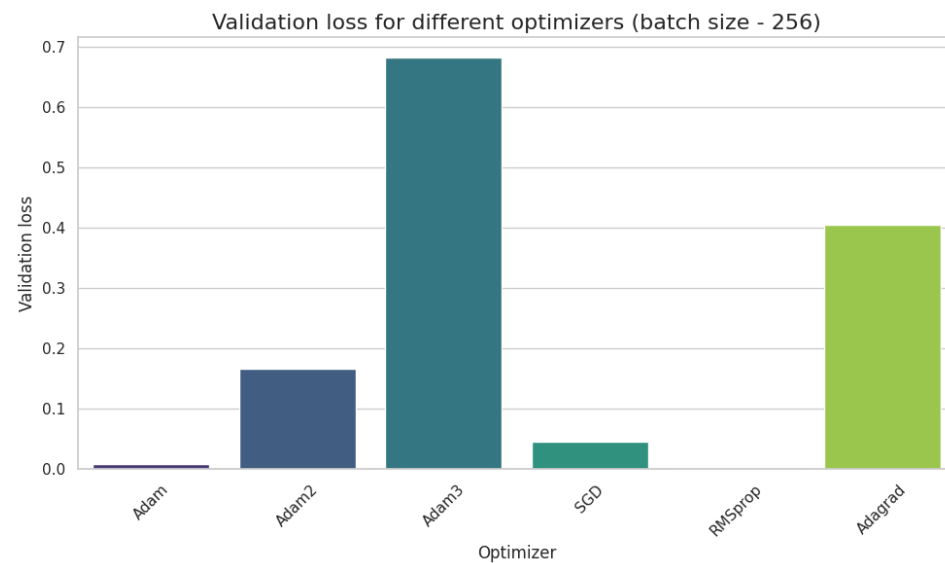
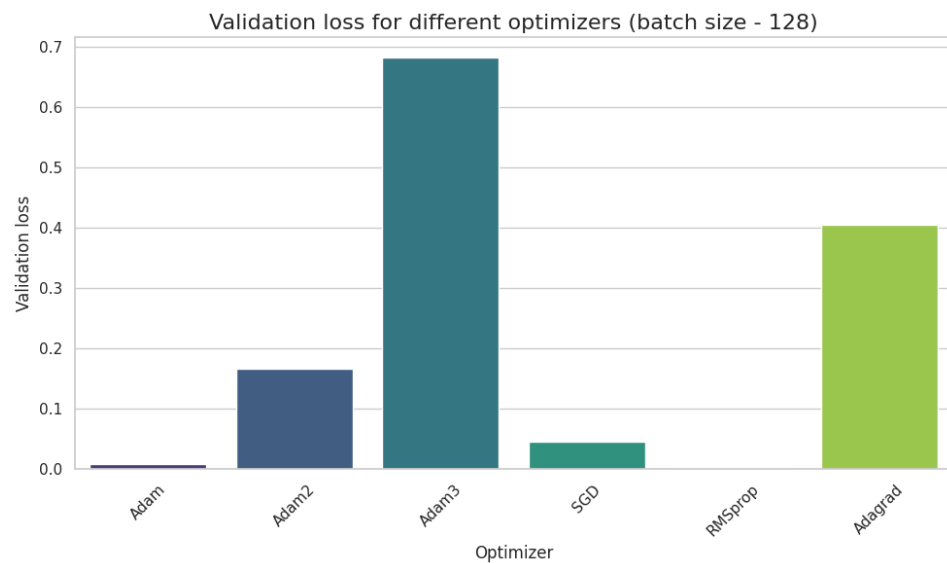
“Both of the tuning parameters β_1 and β_2 are typically set to be close to 1, and common values are $\beta_1 = 0.9$. $\beta_2 = 0.999$. The reason is simply that too small values will effectively result in an exponential forgetting of the past information and remove the – often very valuable – memory effect inherent in this method.”

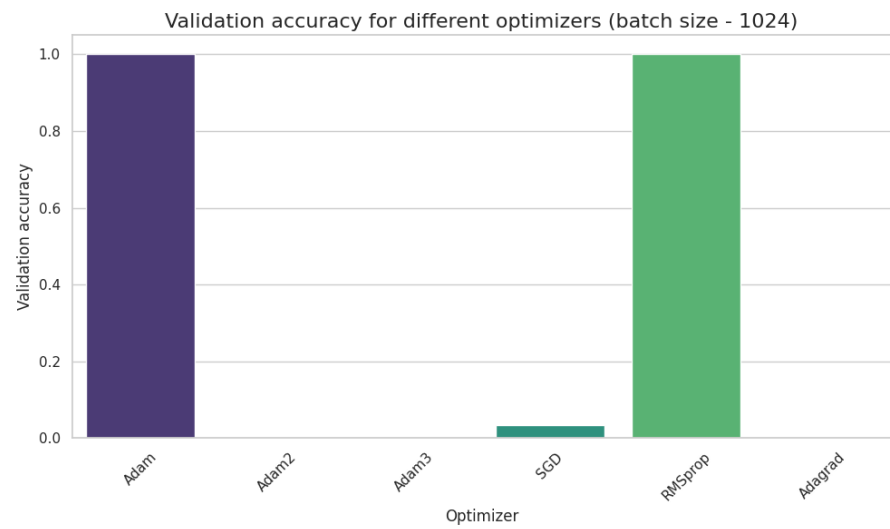
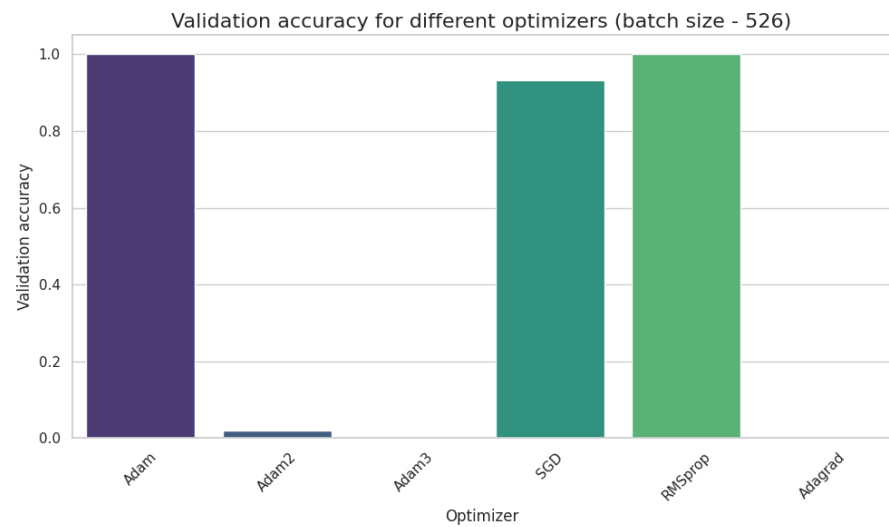
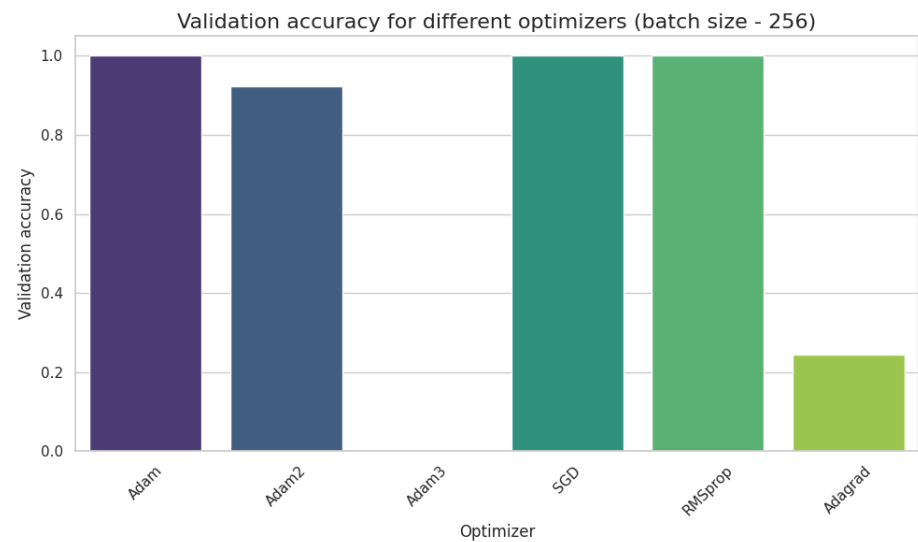
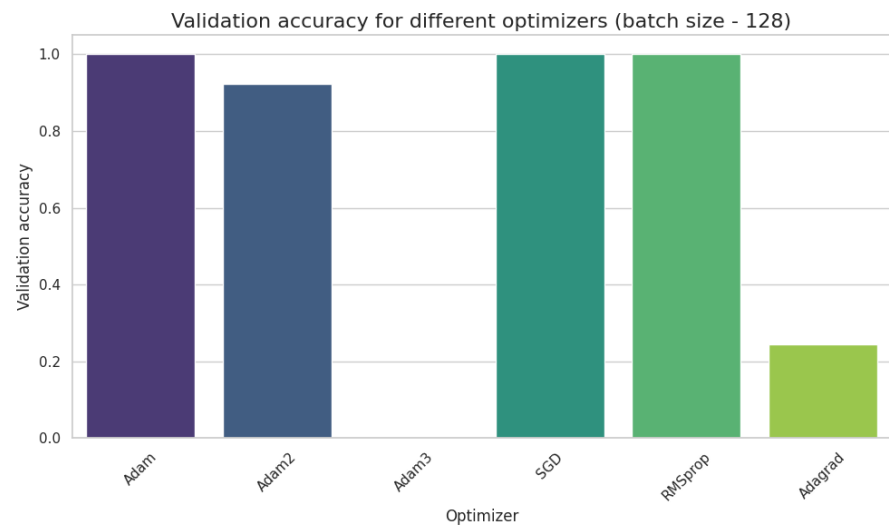
$$d_t = (1 - \beta_1) \sum_{i=1}^t \beta_1^{t-i} \nabla J_i,$$
$$\gamma_t = \frac{\eta}{\sqrt{t}} \left((1 - \beta_2) \text{diag} \left(\sum_{i=1}^t \beta_2^{t-i} \|\nabla J_i\|^2 \right) \right)^{1/2}.$$

OPTIMIZERS

- 1) Adam: Adam(learning_rate=0.001),
- 2) Adam2: Adam(learning_rate=0.0001),
- 3) Adam3: Adam(learning_rate=0.00001),
- 4) SGD: SGD(learning_rate=0.01, momentum=0.9),
- 5) RMSprop: RMSprop(learning_rate=0.001),
- 6) Adagrad: Adagrad(learning_rate=0.01)







Key takeaway:

One could watch a fire burn or water flow forever, and... well, I guess we could add epochs calculated to that list.



THANK YOU VERY MUCH FOR YOUR ATTENTION!